

Experiment 1

SQL Data Definition and Data Manipulation Operations Oracle Database 21c XE

1. SQL Source Code

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-- =====
-- SRIT R23 | Database Management Systems Lab
-- Experiment 1 : SQL Data Definition and Data Manipulation
-- Oracle Database 21c XE
-- =====

SET SERVEROUTPUT ON

-- -----
-- EXPERIMENT 1a : SQL DDL commands (tables WITHOUT constraints)
-- -----

-- 1a-1. Create the tables of the database without constraints
CREATE TABLE student (
    name          VARCHAR2(20),
    student_number NUMBER(3),
    class         NUMBER(1),
    major         VARCHAR2(10)
);
CREATE TABLE course (
    course_name  VARCHAR2(30),
    course_number VARCHAR2(15),
    credit_hours NUMBER(2),
    department   VARCHAR2(15)
);
CREATE TABLE section (
    section_identifier NUMBER(4),
    course_number      VARCHAR2(15),
    semester           VARCHAR2(10),
    year              VARCHAR2(4),
    instructor         VARCHAR2(15)
);
CREATE TABLE grade_report (
    student_number      NUMBER(3),
    section_identifier  NUMBER(4),
    grade              VARCHAR2(5)
);
CREATE TABLE prerequisite (
    course_number      VARCHAR2(15),
    prerequisite_number VARCHAR2(20)
);

-- 1a-2. Insert all values into the tables
INSERT INTO student VALUES ('Smith', 17, 1, 'CS');
INSERT INTO student VALUES ('Brown', 8, 2, 'CS');
INSERT INTO course VALUES ('Intro to Computer Science', 'CS1310', 4, 'CS');
INSERT INTO course VALUES ('Data Structures', 'CS3320', 4, 'CS');
INSERT INTO course VALUES ('Discrete Mathematics', 'MATH2410', 3, 'MATH');
INSERT INTO course VALUES ('Database', 'CS3380', 3, 'CS');
INSERT INTO section VALUES (85, 'MATH2410', 'Fall', '07', 'King');
INSERT INTO section VALUES (92, 'CS1310', 'Fall', '07', 'Anderson');
INSERT INTO section VALUES (102, 'CS3320', 'Spring', '08', 'Knuth');
INSERT INTO section VALUES (112, 'MATH2410', 'Fall', '08', 'Chang');
INSERT INTO section VALUES (119, 'CS1310', 'Fall', '08', 'Anderson');
INSERT INTO section VALUES (135, 'CS3380', 'Fall', '08', 'Stone');
INSERT INTO grade_report VALUES (17, 112, 'B');
INSERT INTO grade_report VALUES (17, 119, 'C');
INSERT INTO grade_report VALUES (8, 85, 'A');
INSERT INTO grade_report VALUES (8, 92, 'A');
INSERT INTO grade_report VALUES (8, 102, 'B');
INSERT INTO grade_report VALUES (8, 135, 'A');
INSERT INTO prerequisite VALUES ('CS3380', 'CS3320');
INSERT INTO prerequisite VALUES ('CS3380', 'MATH2410');
INSERT INTO prerequisite VALUES ('CS3320', 'CS1310');
COMMIT;
```

```

-- 1a-3. Describe all tables
DESC student;
DESC course;
DESC section;
DESC grade_report;
DESC prerequisite;

-- 1a-4. List the created tables
SELECT table_name FROM user_tables ORDER BY table_name;

-- 1a-5. Display the values of each table
SELECT * FROM student;
SELECT * FROM course;
SELECT * FROM section;
SELECT * FROM grade_report;
SELECT * FROM prerequisite;

-- 1a-6. Delete all tables
DROP TABLE grade_report PURGE;
DROP TABLE prerequisite PURGE;
DROP TABLE section PURGE;
DROP TABLE course PURGE;
DROP TABLE student PURGE;

-- -----
-- EXPERIMENT 1b : SQL DML commands (tables WITH constraints)
-- -----

-- 1b-1. Implement the tables using the given constraints
CREATE TABLE student (
    name          VARCHAR2(20),
    student_number NUMBER(3),
    class         NUMBER(1),
    major         VARCHAR2(10) NOT NULL,
    CONSTRAINT pk_student PRIMARY KEY (student_number)
);
CREATE TABLE course (
    course_name   VARCHAR2(30),
    course_number VARCHAR2(15),
    credit_hours  NUMBER(2) NOT NULL,
    department    VARCHAR2(15),
    CONSTRAINT pk_course PRIMARY KEY (course_number)
);
CREATE TABLE section (
    section_identifier NUMBER(4),
    course_number      VARCHAR2(15),
    semester           VARCHAR2(10) NOT NULL,
    year               VARCHAR2(4),
    instructor         VARCHAR2(15),
    CONSTRAINT pk_section PRIMARY KEY (section_identifier)
);
CREATE TABLE grade_report (
    student_number    NUMBER(3),
    section_identifier NUMBER(4),
    grade             VARCHAR2(5) NOT NULL,
    CONSTRAINT pk_grade_report PRIMARY KEY (student_number, section_identifier),
    CONSTRAINT fk_gr_student FOREIGN KEY (student_number)
        REFERENCES student(student_number) ON DELETE CASCADE,
    CONSTRAINT fk_gr_section FOREIGN KEY (section_identifier)
        REFERENCES section(section_identifier) ON DELETE CASCADE
);
CREATE TABLE prerequisite (
    course_number      VARCHAR2(15),
    prerequisite_number VARCHAR2(20),
    CONSTRAINT pk_prerequisite PRIMARY KEY (course_number, prerequisite_number),
    CONSTRAINT fk_prereq_course FOREIGN KEY (course_number)
        REFERENCES course(course_number) ON DELETE CASCADE
);

-- 1b-2. Display the description of each table
DESC student;
DESC course;
DESC section;
DESC grade_report;
DESC prerequisite;
-- 1b-3. Insert the values of the database

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```

INSERT INTO student VALUES ('Smith', 17, 1, 'CS');
INSERT INTO student VALUES ('Brown', 8, 2, 'CS');
INSERT INTO course VALUES ('Intro to Computer Science', 'CS1310', 4, 'CS');
INSERT INTO course VALUES ('Data Structures', 'CS3320', 4, 'CS');
INSERT INTO course VALUES ('Discrete Mathematics', 'MATH2410', 3, 'MATH');
INSERT INTO course VALUES ('Database', 'CS3380', 3, 'CS');
INSERT INTO section VALUES (85, 'MATH2410', 'Fall', '07', 'King');
INSERT INTO section VALUES (92, 'CS1310', 'Fall', '07', 'Anderson');
INSERT INTO section VALUES (102, 'CS3320', 'Spring', '08', 'Knuth');
INSERT INTO section VALUES (112, 'MATH2410', 'Fall', '08', 'Chang');
INSERT INTO section VALUES (119, 'CS1310', 'Fall', '08', 'Anderson');
INSERT INTO section VALUES (135, 'CS3380', 'Fall', '08', 'Stone');
INSERT INTO grade_report VALUES (17, 112, 'B');
INSERT INTO grade_report VALUES (17, 119, 'C');
INSERT INTO grade_report VALUES (8, 85, 'A');
INSERT INTO grade_report VALUES (8, 92, 'A');
INSERT INTO grade_report VALUES (8, 102, 'B');
INSERT INTO grade_report VALUES (8, 135, 'A');
INSERT INTO prerequisite VALUES ('CS3380', 'CS3320');
INSERT INTO prerequisite VALUES ('CS3380', 'MATH2410');
INSERT INTO prerequisite VALUES ('CS3320', 'CS1310');
COMMIT;

-- 1b-4. Display the instances of each table
SELECT * FROM student;
SELECT * FROM course;
SELECT * FROM section;
SELECT * FROM grade_report;
SELECT * FROM prerequisite;

-- 1b-5. Add a branch attribute to student and describe the table
ALTER TABLE student ADD branch VARCHAR2(10);
DESC student;

-- 1b-6. Copy major values into branch and display it
UPDATE student SET branch = major;
SELECT * FROM student;

-- 1b-7. Remove the major attribute from student
ALTER TABLE student DROP COLUMN major;
DESC student;

-- 1b-8. Rename course_number to cid in course and describe it
ALTER TABLE course RENAME COLUMN course_number TO cid;
DESC course;

-- 1b-9. Change the credit hours of all courses to 4
UPDATE course SET credit_hours = 4;
SELECT * FROM course;

-- 1b-10. Put a NOT NULL constraint on branch in student
ALTER TABLE student MODIFY branch NOT NULL;
DESC student;

-- 1b-11. Rename the student table to pupil
ALTER TABLE student RENAME TO pupil;

-- 1b-12. Remove the student table (it is now named pupil, so this gives an error)
DROP TABLE student;

-- 1b-13. Remove the rows of the 'Fall' semester in section
DELETE FROM section WHERE semester = 'Fall';
SELECT * FROM section;

-- 1b-14. Remove the row of 'Data Structures' in course
DELETE FROM course WHERE course_name = 'Data Structures';
SELECT * FROM course;
COMMIT;

-- 1b-15. Remove all rows of all tables using TRUNCATE
TRUNCATE TABLE grade_report;
TRUNCATE TABLE prerequisite;
TRUNCATE TABLE pupil CASCADE;
TRUNCATE TABLE section CASCADE;
TRUNCATE TABLE course CASCADE;

-- 1b-16. Remove pupil, course and section so that they stay in the recycle bin

```

```
DROP TABLE pupil CASCADE CONSTRAINTS;  
DROP TABLE course CASCADE CONSTRAINTS;  
DROP TABLE section CASCADE CONSTRAINTS;  
SHOW RECYCLEBIN;  
  
-- 1b-17. Remove grade_report and prerequisite tables permanently  
DROP TABLE grade_report PURGE;  
DROP TABLE prerequisite PURGE;  
SELECT table_name FROM user_tables;
```

2. Execution and Output


```
SQL*Plus - Output

[1a-3]
Name                                     Null?      Type
-----
NAME                                     VARCHAR2(20)
STUDENT_NUMBER                           NUMBER(3)
CLASS                                    NUMBER(1)
MAJOR                                    VARCHAR2(10)

Name                                     Null?      Type
-----
COURSE_NAME                             VARCHAR2(30)
COURSE_NUMBER                           VARCHAR2(15)
CREDIT_HOURS                             NUMBER(2)
DEPARTMENT                              VARCHAR2(15)

Name                                     Null?      Type
-----
SECTION_IDENTIFIER                       NUMBER(4)
COURSE_NUMBER                           VARCHAR2(15)
SEMESTER                                VARCHAR2(10)
YEAR                                    VARCHAR2(4)
INSTRUCTOR                              VARCHAR2(15)

Name                                     Null?      Type
-----
STUDENT_NUMBER                           NUMBER(3)
SECTION_IDENTIFIER                       NUMBER(4)
GRADE                                    VARCHAR2(5)

Name                                     Null?      Type
-----
COURSE_NUMBER                           VARCHAR2(15)
PREREQUISITE_NUMBER                     VARCHAR2(20)

[1a-4]
TABLE_NAME
-----
COURSE
GRADE_REPORT
PREREQUISITE
SECTION
STUDENT

5 rows selected.

Rendered from expl_output.txt (sample output)
```

[1a-5]

NAME	STUDENT_NUMBER	CLASS	MAJOR
Smith	17	1	CS
Brown	8	2	CS

2 rows selected.

COURSE_NAME	COURSE_NUMBER	CREDIT_HOURS	DEPARTMENT
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

4 rows selected.

SECTION_IDENTIFIER	COURSE_NUMBER	SEMESTER	YEAR	INSTRUCTOR
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

6 rows selected.

STUDENT_NUMBER	SECTION_IDENTIFIER	GRADE
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

6 rows selected.

COURSE_NUMBER	PREREQUISITE_NUMBER
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

3 rows selected.

[1a-6]

Table dropped.

Table dropped.

Table dropped.

Table dropped.

Table dropped.

Rendered from exp1_output.txt (sample output)

```
SQL*Plus - Output

[1b-1]
Table created.

Table created.

Table created.

Table created.

Table created.

[1b-2]
Name                                         Null?    Type
-----
NAME                                         VARCHAR2(20)
STUDENT_NUMBER                             NOT NULL NUMBER(3)
CLASS                                       NUMBER(1)
MAJOR                                       NOT NULL VARCHAR2(10)

Name                                         Null?    Type
-----
COURSE_NAME                                VARCHAR2(30)
COURSE_NUMBER                             NOT NULL VARCHAR2(15)
CREDIT_HOURS                              NOT NULL NUMBER(2)
DEPARTMENT                                VARCHAR2(15)

Name                                         Null?    Type
-----
SECTION_IDENTIFIER                         NOT NULL NUMBER(4)
COURSE_NUMBER                             VARCHAR2(15)
SEMESTER                                  NOT NULL VARCHAR2(10)
YEAR                                       VARCHAR2(4)
INSTRUCTOR                                VARCHAR2(15)

Name                                         Null?    Type
-----
STUDENT_NUMBER                             NOT NULL NUMBER(3)
SECTION_IDENTIFIER                         NOT NULL NUMBER(4)
GRADE                                       NOT NULL VARCHAR2(5)

Name                                         Null?    Type
-----
COURSE_NUMBER                             NOT NULL VARCHAR2(15)
PREREQUISITE_NUMBER                       NOT NULL VARCHAR2(20)

Rendered from expl_output.txt (sample output)
```

[1b-3]

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

1 row created.

Commit complete.

Rendered from exp1_output.txt (sample output)

[1b-4]

NAME	STUDENT_NUMBER	CLASS	MAJOR
------	----------------	-------	-------

Smith	17	1	CS
Brown	8	2	CS

2 rows selected.

COURSE_NAME	COURSE_NUMBER	CREDIT_HOURS	DEPARTMENT
-------------	---------------	--------------	------------

Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

4 rows selected.

SECTION_IDENTIFIER	COURSE_NUMBER	SEMESTER	YEAR	INSTRUCTOR
--------------------	---------------	----------	------	------------

85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

6 rows selected.

STUDENT_NUMBER	SECTION_IDENTIFIER	GRADE
----------------	--------------------	-------

17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

6 rows selected.

COURSE_NUMBER	PREREQUISITE_NUMBER
---------------	---------------------

CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

3 rows selected.

[1b-5]

Table altered.

Name	Null?	Type
------	-------	------

NAME		VARCHAR2(20)
STUDENT_NUMBER	NOT NULL	NUMBER(3)
CLASS		NUMBER(1)
MAJOR	NOT NULL	VARCHAR2(10)
BRANCH		VARCHAR2(10)

Rendered from exp1_output.txt (sample output)

```
SQL*Plus - Output

[1b-6]
2 rows updated.

NAME                STUDENT_NUMBER CLASS MAJOR      BRANCH
-----
Smith                17          1 CS        CS
Brown                8           2 CS        CS

2 rows selected.

[1b-7]
Table altered.

Name                Null?   Type
-----
NAME                NOT NULL VARCHAR2(20)
STUDENT_NUMBER      NOT NULL NUMBER(3)
CLASS               NOT NULL NUMBER(1)
BRANCH              NOT NULL VARCHAR2(10)

[1b-8]
Table altered.

Name                Null?   Type
-----
COURSE_NAME         NOT NULL VARCHAR2(30)
CID                 NOT NULL VARCHAR2(15)
CREDIT_HOURS        NOT NULL NUMBER(2)
DEPARTMENT          NOT NULL VARCHAR2(15)

[1b-9]
4 rows updated.

COURSE_NAME          CID          CREDIT_HOURS DEPARTMENT
-----
Intro to Computer Science CS1310          4 CS
Data Structures      CS3320          4 CS
Discrete Mathematics MATH2410        4 MATH
Database             CS3380          4 CS

4 rows selected.

[1b-10]
Table altered.

Name                Null?   Type
-----
NAME                NOT NULL VARCHAR2(20)
STUDENT_NUMBER      NOT NULL NUMBER(3)
CLASS               NOT NULL NUMBER(1)
BRANCH              NOT NULL VARCHAR2(10)

[1b-11]
Table altered.

[1b-12]
DROP TABLE student
*
ERROR at line 1:
ORA-00942: table or view does not exist

Rendered from expl_output.txt (sample output)
```

```
SQL*Plus - Output

[1b-13]
5 rows deleted.

SECTION_IDENTIFIER COURSE_NUMBER SEMESTER YEAR INSTRUCTOR
-----
102 CS3320 Spring 08 Knuth

1 row selected.

[1b-14]
1 row deleted.

COURSE_NAME CID CREDIT_HOURS DEPARTMENT
-----
Intro to Computer Science CS1310 4 CS
Discrete Mathematics MATH2410 4 MATH
Database CS3380 4 CS

3 rows selected.

Commit complete.

[1b-15]
Table truncated.

Table truncated.

Table truncated.

Table truncated.

Table truncated.

[1b-16]
Table dropped.

Table dropped.

Table dropped.

ORIGINAL NAME RECYCLEBIN NAME OBJECT TYPE DROP TIME
-----
PUPIL BIN$aB3xK9dQTr2mZ0pLwE5cNg==$0 TABLE 2026-10-02:10:15:21
COURSE BIN$Yq7fH1sVSe+u4RkTz8JdXA==$0 TABLE 2026-10-02:10:15:21
SECTION BIN$m2LcP6wUQJ6hD3oBn0vYtw==$0 TABLE 2026-10-02:10:15:21

[1b-17]
Table dropped.

Table dropped.

no rows selected

Rendered from expl_output.txt (sample output)
```